Haitham Abu Khaled

Backend & AI Engineer · Computer Science student, PPU

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Fourth-year Computer Science student (92.4%, President's Honor List five times) building backend systems where AI interprets messy input and deterministic rules make the decision. Shipping a production AI feature in Java and Spring Boot as a backend intern, and team lead of the 1st-place team at Ainak Challenges 2026. Interested in backend engineering, applied AI and distributed systems.

◇ Experience

Backend Development Intern · Co.Te.De Technologies, Hebron — Jul 2026 – Present

- ◇ Built the backend of “Note to Action”, an AI assistant for a multi-tenant debt-management system (Java, Spring Boot, PostgreSQL) that turns merchants' typed or spoken notes in Arabic or English into proposed ledger actions, applied only after the merchant confirms.
- ◇ Designed an asynchronous accept-then-poll pipeline combining OpenAI speech-to-text with schema-constrained LLM extraction and Arabic-aware name and amount parsing.
- ◇ Found and fixed money-affecting bugs by testing against the live model and database, including amounts misread by a factor of 1,000 and a reminder that would have reached every overdue customer.
- ◇ Wrote ~700 automated tests (~90% branch coverage, JaCoCo) and Liquibase migrations for dedicated and shared row-level-secured tenant databases; recorded 26 design decisions and owned the API contract with the Flutter developer.

Video Pipeline Developer (Remote) · Verbivore, via Gaza Sky Geeks “Train to Hire” — May – Aug 2026

- ◇ Prototyped a CSV- and template-driven pipeline (Python, Creatomate, ffmpeg, DaVinci Resolve) that turns bilingual Arabic–English lesson scripts into course videos; built and tested end to end, rendering 495 clips across 12 units.
- ◇ Fixed mixed right-to-left text corruption with Unicode directional isolates, and traced a ~2.5× render-cost overrun to a 60 fps template default, confirmed with vendor support (moving to 25 fps cuts cost ~58%).
- ◇ Prototyped a text-to-speech pipeline on Amazon Polly (Arabic neural voice, custom pronunciation lexicon, S3 storage) and audited all 12 source scripts for consistency and language errors.

QA Intern (Training Program) · Ahl Logics, Hebron — Jul – Nov 2024

- ◇ Completed hands-on training in automated software testing with Python and Selenium.

◇ Projects

MobilityBridge · Team lead, 1st place at Ainak Challenges 2026 — Jul 2026 – Present

- ◇ Led a team of three to 1st place at the Arab Innovation Network's hackathon (Birzeit University) with a platform that helps Palestinian students find and assess international mobility opportunities.
- ◇ Validated the problem with a survey of 133 students across 7 universities; designed a deterministic, domain-agnostic eligibility engine where AI explains and rules decide. Presented the live prototype and year-one budget.

React · TypeScript · Spring Boot · PostgreSQL · Claude API

Laqeetha · Graduation project, team of three — 2026 – 2027

- ◇ AI-assisted lost-and-found platform for the PPU campus: photo tagging, lost/found matching and ownership verification before handover. Selected for the Palestine XI National Forum for Creativity and Excellence (Oct 2026).

SmartCity Data Layer · Group project, Non-Relational Databases — 2026

- ◇ MongoDB store for city service requests with query-driven indexes and geo data, synchronised to Elasticsearch through Change Streams for full-text search.

◇ Education

BSc Computer Science · Palestine Polytechnic University, Hebron — Sep 2023 – Jun 2027 (expected)

Cumulative 92.4%, major 92.9%; President's Honor List for five semesters. Coursework: Data Structures, Algorithm Design & Analysis, Database Systems, Non-relational Databases, Computer Networks, Principles of AI, Programming Languages, Theory of Computing.

Erasmus+ Exchange Semester, Computer Science · Dunărea de Jos University of Galați, Romania — Oct 2025 – Feb 2026

PPU's sole exchange student that semester. Average 9.25/10: Web Programming 10, Image Processing 10, Computer-Aided Graphics 9, Operating Systems 8.

◇ Honors & Awards

1st place, Ainak Challenges 2026 (Arab Innovation Network) — Aug 2026

Certificate of Pride, PPU President — Aug 2026

1st place, Fikra La Bukra camp (Qaddumi Foundation) — Aug 2025

Dual scholarship, Qaddumi Foundation & Reach Education Fund — 2023 – Present

Presenter, Palestine XI National Forum — Oct 2026

PPU delegate, Mizan 3 e-justice launch — Sep 2026

3rd place in Palestine, IEEE preXtreme — Sep 2024

President's Honor List, five semesters — 2023 – 2026

◇ Leadership & Programs

Palestinian Youth Leaders Program, Amideast (100 hours) — 2025

IEEE X-Camp, leadership & soft-skills camp — 2024

Soliya Connect Bel3araby, virtual exchange — 2026

Booth representative, Softofy Tech at LCOY Palestine — 2026

◇ Skills

Languages	Java, Python, JavaScript / TypeScript, C++, SQL
Backend	Spring Boot, JPA/Hibernate, REST API design, Node.js / Express, Liquibase, JUnit, JaCoCo
Data	PostgreSQL, MongoDB, Elasticsearch
AI	OpenAI and Claude APIs, speech-to-text, schema-constrained LLM extraction, OpenCV
Tools	Git, Docker, Linux, AWS (Polly, S3), ffmpeg, Selenium, Postman
Spoken	Arabic (native), English (proficient; EF SET 76/100, C2)